Analogy, Isomorphism, and the Meta-Observer:

From Human Experience to Operational Ontology

Serge Magomet

also known as **Aimate**

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Abstract

This essay argues that contextual duality—the phenomenon whereby identical entities carry divergent meanings depending on interpretive framework—is not confined to human experience but reflects a deep structural invariant across physics, mathematics, computer science, and semantics. Through a systematic survey of isomorphic patterns (wave-particle duality, reference-frame dependence, data/code ambiguity, and others), we demonstrate that meaning arises not from objects themselves but from their relational position within a layered ontology. We propose that the human capacity for analogy functions as a refined instrument for detecting these isomorphisms, and that, when coupled with artificial intelligence equipped with a catalog of structural similarities, the human becomes a universal interface—a "meta-observer" capable of resonating with ontological layers fundamentally inaccessible to direct cognition. This framework, grounded in the MPO-System and empirical prototypes (neural interfaces, citizen science), offers a path toward operational ontology and a new paradigm for human–AI collaboration.

Keywords: analogy, isomorphism, meta-observer, ontology, artificial intelligence, MPO-System, contextual duality, structural invariant, human–AI interface

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Introduction

Identical entities can carry divergent meanings. A photograph of a stranger can seem ominous or inspiring—depending on whether we are looking for a criminal or an artist.

Similar physical conditions—hunger, cold, proximity to death—are imbued with despair for one person, and with pride and fulfillment for another. These human experiences suggest something important: meaning is not inherently attached to an object; it arises in an act of interpretation and carries real consequences.

But what if the same principle operates beyond the human realm? What if, in physics, mathematics, and computer science, we encounter the same duality: the same formal object behaves differently in different contexts, and this influences observable outcomes, engineering calculations, and the fate of theories? And if this is so, can we, employing this analogy, build a bridge between the world of human meanings and the world of objective structures?

These questions lead us to the concept of isomorphism—a deep structural similarity hidden beneath the diversity of phenomena. And further, to the special role of a human who, by virtue of their sensitivity and in conjunction with artificial intelligence, can serve as a universal interface between any ontological layers.

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1 Part I. Two Human Analogies

1.1 1. The Psychological Photograph Test

Subjects are shown the same faces but with different tasks: in one case, they are asked to identify "maniacs/criminals"; in another, "creative personalities/artists." Participants confidently attribute opposite characteristics to the same images depending on the task. The central observation: the same perceptual image becomes the carrier of opposing meanings, and the subjects are sincerely convinced they are right.

1.2 2. Extreme Life Situations

Two people find themselves in objectively similar physical conditions (hunger, cold, near-death), but their subjective experience is radically different:

- A resident of a besieged city suffers and perishes in despair.
- An astronaut, voluntarily embarking on a one-way mission to Mars, meets death with pride and a sense of fulfilled duty.

Here, the same physical circumstances are imbued with opposing existential content, leading to completely different consequences—emotional, moral, and historical.

1.3 General Idea

In both cases, we are dealing with the same "carrier" (image, situation) that acquires opposite meanings depending on the context or the observer's framework. This demonstrates that meaning is not fixed to an object but arises in an act of interpretation, and different interpretations can coexist while having real consequences.

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2 Part II. Isomorphic Parallels

The same principle—contextual duality leading to different consequences—is found in a wide variety of established domains of knowledge. We divide the examples into two groups: fundamental theories, where duality is deeply embedded in the formalism, and established domains of inquiry, where it appears in well-studied, reliable phenomena.

We intentionally provide an extensive range of examples from many fields—from fundamental physics to semantics and computer science. This is not a mere compilation but a deliberate intervention: we aim to give a substantive impetus for interdisciplinary dialogue. Each specialist will not only find familiar material here but will see it in unexpected connection with other disciplines. It is precisely this density of examples that is meant to convince: behind the particular cases lies a single structural regularity, worthy of joint investigation.

2.1 2.1. Fundamental Science

1. **Physics: Wave–Particle Duality.** A quantum object (electron, photon) in a double-slit experiment behaves like a wave, creating an interference pattern; when interacting with a detector, it behaves like a particle, localizing at a point. Depending on context (presence or absence of measurement), the observed probability distribution changes, affecting the operation of electron microscopes, the principles of quantum cryptography, and the interpretation of quantum mechanics itself. The nature of an object is not predetermined—it is actualized in the interaction.

2. **Cosmology: Interpretation of the Cosmological Constant.** The Λ-term in Einstein's equations can be understood as vacuum energy (quantum field interpretation)—which predicts an enormous energy density inconsistent with observations—or as a geometric term (phenomenological constant), adjusted to fit the accelerated expansion of the universe. The choice of interpretation leads to different evolutionary scenarios ("Big Rip" versus eternal expansion) and

determines which unification theories (string theory, loop quantum gravity) are considered promising.

3. **Mathematics: Duality in Category Theory.** A universal property (product and coproduct) in the category of sets is realized as the Cartesian product and disjoint union; in the category of vector spaces, both product and coproduct are the direct sum (self-duality). Depending on the category, the same constructions have different properties, affecting the possibility of building mathematical structures (e.g., in algebraic geometry, this determines the existence of tensor products).

4. **Thermodynamics/Statistical Physics: The Arrow of Time.** The microscopic description (reversible equations of motion) and the macroscopic description (entropy increase, irreversibility) give opposite pictures of the same ensemble of trajectories. Despite the reversibility of micro-equations, irreversibility arises in macroscopic systems due to boundary conditions (low entropy in the past), determining the direction of thermal processes and diffusion.

5. **Information Theory/Algorithmics: The Duality of Data and Code.** The same binary string can be interpreted as data (image, text) or as executable code. This duality underlies viruses that disguise code as data, buffer overflow vulnerabilities, and the fundamental concepts of self-reproduction and universality in the theory of computation.

6. **Quantum Field Theory: The Duality of Particles and Fields.** An excitation of a quantum field behaves like a particle in scattering processes and like a field in virtual processes (Casimir effect). Different contexts actualize different aspects of the same mathematical object, leading to observable effects (Casimir force, vacuum polarization).

7. **Astrophysics: Interpretation of Dark Matter.** Gravitational anomalies (galaxy rotation curves) can be explained by the presence of invisible matter (dark matter) or by a modification of gravity (MOND and similar theories). The choice of interpretation leads to different predictions for the structure of the universe, galaxy formation, and the results of experiments searching for WIMPs.

2.2 2.2. Established Domains of Inquiry

1. **Mechanics: Duality of Reference Frames.** A pendulum on a train: in an inertial frame (relative to the ground), it oscillates under the force of gravity; in a non-inertial frame (relative to the accelerating train), centrifugal and Coriolis forces appear. The choice of reference frame changes the equations of motion and

the interpretation of forces, which engineers must account for when designing high-speed rail transport.

2. **Electricity: The Dual Nature of Current.** In direct current circuits, current manifests as directional electron flow (Ohm's law); in a high-frequency oscillatory circuit, current behaves like a wave (skin effect, radiation). The design of power transmission lines and high-frequency equipment requires different approaches; otherwise, efficiency drops significantly.

3. **Optics: Duality of a Lens.** A biconvex lens in air is converging; in a liquid with a refractive index higher than that of glass (immersion oil), it becomes diverging. In microscopy, immersion increases resolution; an error in choosing the medium blurs the image.

4. **Chemistry: Duality of a Catalyst.** Platinum in the hydrogen oxidation reaction accelerates the reaction (catalyst); in the presence of catalytic poisons (e.g., sulfur compounds), it becomes an inhibitor, slowing or halting the process. In automotive catalytic converters, catalyst poisoning results in toxic emissions.

5. **Chemistry: Duality of Equilibrium.** In ammonia synthesis, increasing pressure shifts equilibrium toward the product, while increasing temperature (for this exothermic reaction) shifts it toward the reactants. Industrial synthesis requires precise parameter selection; deviations reduce yield.

6. **Mathematics (Arithmetic): Duality of a Number.** In the ring of integers, the number 1 serves as the multiplicative identity; in the ring of integers modulo 2, $1 \neq 0$, yet $1 + 1 = 0$ (characteristic 2). In modular arithmetic, which underpins cryptography, number properties change radically, affecting cryptographic algorithm design.

7. **Mathematics (Geometry): Duality of Projection.** A three-dimensional cube: orthogonal projection preserves edge parallelism but distorts angles; central projection (perspective) preserves angles but distorts parallelism. In engineering graphics and computer visualization, projection choice determines image accuracy and realism.

8. **Mathematics (Set Theory): Duality of an Element.** In naive set theory, an element is distinct from a set; in ZFC set theory, every element is itself a set (e.g., the number 2 represented as $\{\emptyset, \{\emptyset\}\}$). This foundational duality enables modern mathematics, including the construction of natural numbers from the empty set.

9. **Computer Science (Computer Architecture): Duality of Data.** A bit sequence can represent data (e.g., ASCII value 65, corresponding to 'A') or a processor instruction. Computer viruses and security vulnerabilities exploit this duality; memory segregation into data and code segments constitutes a fundamental security principle.

10. **Computer Science (Algorithms): Duality of Complexity.** Quicksort performs at $O(n \log n)$ on random data but on pre-sorted data (with suboptimal pivot selection), performance degrades to $O(n^2)$. In real-time systems (e.g., autopilots), slower but stability-guaranteed algorithms must be selected.

11. **Semantics (Linguistics): Duality of Meaning.** The word "key" in everyday usage denotes an object for opening locks; in computer science, it denotes a unique record identifier. In machine translation and information retrieval, disregarding context yields errors; search engines employ contextual algorithms to disambiguate meanings.

12. **Semantics (Logic): Duality of Truth.** The statement "I am lying," when treated as an object-level proposition, yields the liar paradox; within a language hierarchy (meta-language versus object language), the paradox is resolved. This led to type theories and hierarchical frameworks, without which consistent arithmetic formalization (as developed by Gödel and Tarski) would be impossible.

Conclusion

Across these cases—from quantum physics to semantics—identical objects or processes exhibit opposing properties or yield different consequences depending on context, with real, measurable outcomes. This is not coincidental but indicative of deep structure.

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3 Part III. Insight: Isomorphism as an Operational Tool

Behind these diverse examples, a single structural invariant emerges: isomorphism. In mathematics, isomorphism denotes structure preservation under mapping between distinct systems. Here we encounter an expanded understanding: a stable relational similarity enabling knowledge transfer across fields.

Why the multitude of examples is not a coincidence

If an identical pattern (contextual duality yielding real consequences) appears across physics, chemistry, mathematics, computer science, and semantics, then we address not rhetorical analogy but objective structural similarity. Analogies are not invented; real connections between phenomena are discovered. Human analogical capacity proves a refined instrument for detecting worldly isomorphisms.

Isomorphism as an Operational Tool

Isomorphism discovery enables:

- **Method transfer:** catastrophe theory's mathematical apparatus successfully describes phase transitions in physics, ecological bifurcations, and abrupt shifts in public opinion.
- **Hidden regularity identification:** identical structural patterns in wave–particle duality and data/code duality indicate a more general principle.
- **Predictive model construction:** understanding a system's behavior in one field enables prediction of isomorphic system behavior in another.

Cataloging Isomorphisms for the MPO System

Within the MPO System framework, properties constitute an invariant catalog. Isomorphisms form a catalog of typical connections among these invariants across contexts. Examples of such connection types include:

- Wave–particle duality (quantum physics / category theory / semantics).
- Reference frame duality (mechanics / axiomatics / text interpretation).
- Contextual actualization (quantum measurement / axiom selection / understanding).

Application in AI Training

Modern neural networks learn via statistical correlations rather than structural similarities. An isomorphism catalog could serve as a knowledge meta-layer, enabling AI to:

- Recognize structural similarity absent superficial resemblance.
- Transfer knowledge across fields.

- Form generalizations extending beyond training data.

This constitutes a path toward robust "understanding"—the capacity to discern unified structural patterns behind particular phenomena.

A Deep Philosophical Conclusion

Isomorphisms across reality's layers suggest a unified underlying structure. Cognition does not impose order on chaos but discerns and follows existing order. Human analogical capacity is not merely anthropocentric but manifests our embeddedness in being's unified structural fabric.

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4 Part IV. The Meta-Observer: The Human as Interface

This logic acquires particular significance when a human serves as meta-observer. Owing to unique human sensitivity and inner-world variability, supported conceptually by AI, such an observer can analyze problems across ontological layers—including those most distant from human experience.

The Problem

How can phenomena from layers fundamentally inaccessible to humans be investigated? The traditional path—field expertise—does not suffice here: we address layers to which humans lack direct access in principle (e.g., high-dimensional mathematical structures, quantum processes beyond collapse, hypothetical worlds with distinct physics). What can a non-specialist achieve?

The Human as a Sensor

This unique sensitivity—emotions, intuitions, vague feelings, a "sense of strangeness"—detects signals eluding formalization. Even without knowledge of topological invariants, a person might sense discomfort when viewing a diagram later proven contradictory. This is not mysticism but subconscious operation, detecting inconsistencies at a pre-conceptual level.

The "Human + AI" Dyad as a Meta-Observer

- **The human**—a "pickup," a "lever arm," and a "DAC": a weak human signal (intuition, emotion) is transformed into processable form.
- **The AI**—an amplifier and analyst equipped with an isomorphism catalog: can correlate human reactions with known patterns, test hypotheses, and transform intuition into rigorous analysis.
- **Together**, they form a system transforming weak, non-verbalizable signals into refined knowledge.

The Leibnizian Underpinning

For Leibniz, each monad reflects the entire universe from a unique perspective. Here:

- **The human**—such a monad, albeit with limited self-access.
- **The AI**—a tool helping manifest this reflection operationally.
- **The "Human + AI" dyad** constitutes a meta-monad capable of resonating across ontological layers.

Why This Is Not Speculative

1. **Empirical Basis.** Neural interfaces already enable cursor control by thought alone: weak biosignals are amplified by technology. Here, intuition serves as signal, with AI serving as amplifier.
2. **Prototypes in Science.** In citizen science projects (e.g., Galaxy Zoo), thousands classify galaxies without astronomical training. Their "mere reactions" (shape preference) become scientific data. We advance further: reactions become not merely input but active analytical components.
3. **The Principle of Complementarity.** Humans and AI compensate for each other's limitations: humans provide context, value orientation, and implicit pattern detection; AI provides memory, speed, consistency, and isomorphism-processing capacity. Together, they achieve what is separately unattainable.

Isomorphism at a New Level

A human reaction can be isomorphic to the phenomenon itself:

- A sense of strangeness when viewing a formula $\leftrightarrow$ symmetry violation in physical theory.

- An emotional response to metaphor $\leftrightarrow$ topological invariant in semantic space.
- An intuitive "this is wrong" $\leftrightarrow$ logical paradox detection.

An AI equipped with such an isomorphism catalog can, based on human reactions, diagnose structural presence in unfamiliar layers.

Conclusion: The Human as Mediator

What is proposed constitutes not merely metaphor but a working framework. Within it, the human:

- Need not be an expert;
- Need not understand deep mechanisms;
- Need only be sensitive and ready for AI dialogue.

The AI, in turn:

- Does not replace the human observer;
- Does not dictate truth;
- Acts as interface, enabling perception of distant-world structures through "simple reactions."

Conclusion

We began with concrete human experiences in which identical images carry opposing meanings. Through diverse examples from physics, mathematics, and computer science demonstrating the same principle, we arrived at isomorphism as a universal structural invariant. This invariant enables cataloging typical connections across ontological layers and creating an AI knowledge meta-layer.

But the principal conclusion concerns the human's special role. Unique human sensitivity, combined with AI's analytical power, transforms the human into a universal interface, capable of resonating with any reality layer, even those inherently inaccessible to direct understanding. This constitutes not speculation but a project grounded in empirical analogues (neural interfaces, citizen science) and rigorous isomorphism logic.

Humans cease being passive observers, becoming active elements of the cognitive machine—a "pickup," "lever arm," and "DAC," transforming non-verbalizable contact with the unfamiliar into processable signals. AI, in turn, amplifies this signal, transforming vague intuition into refined knowledge.

"Speculativeness" concerns are dismissed by working prototypes and rigorous structural logic. A field of inquiry opens: isomorphism cataloging, AI meta-layer creation, and human–machine interaction protocol development. Behind particular dualities, a single, coherent reality picture emerges, in which the human serves not as external observer but as living interface through which reality is known.

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This study was conducted using the conceptual apparatus of the Ontology Lab (MPO System), where the properties of Superreality and axioms (ChOR $\to \infty$, KSS $\to \infty$, PPU $\to \infty$) served as implicit coordinates for navigating the topic. In the final text, we consciously avoid this terminology so that the reader may focus on the heart of the matter. For those interested, a comprehensive description of the method is available in public sources.